

A.G.MANOJ

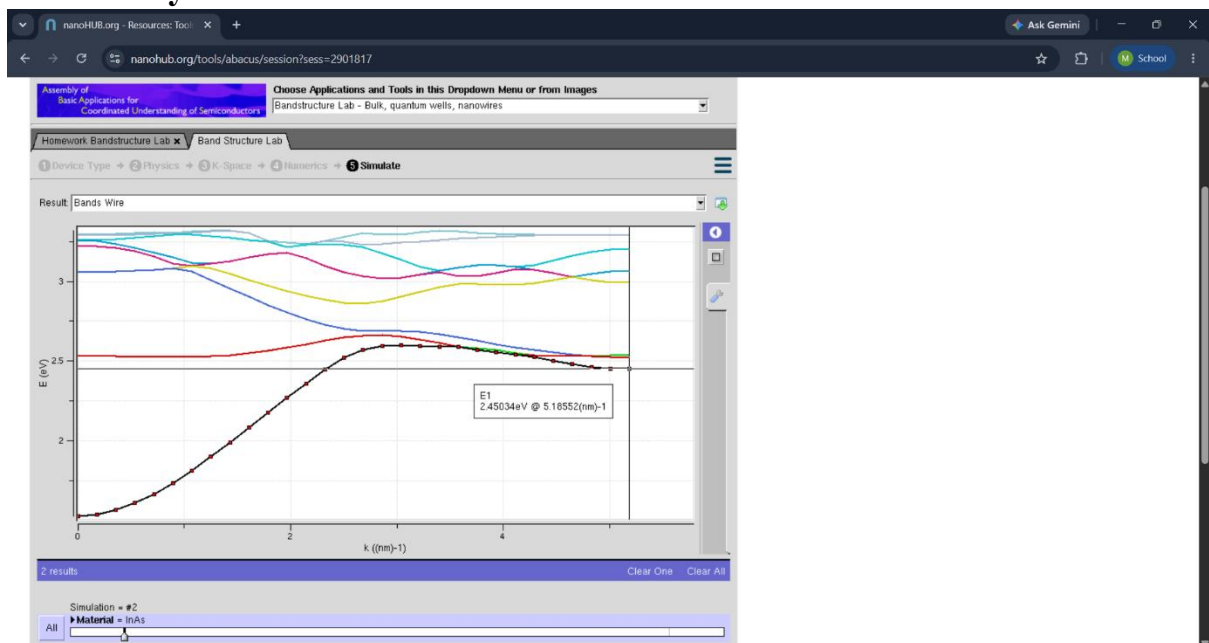
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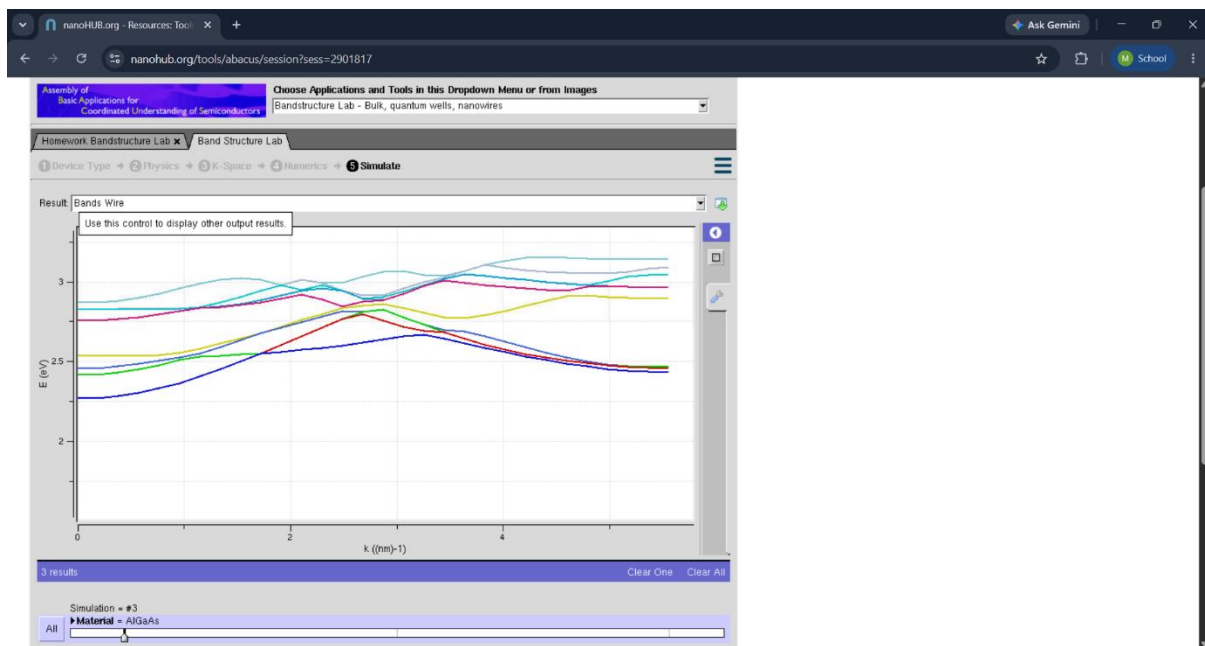
ECA3102-NANO ELECTRONICS

Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub

Case Study 1: Simulate the band-structure of InAs circular nanowire

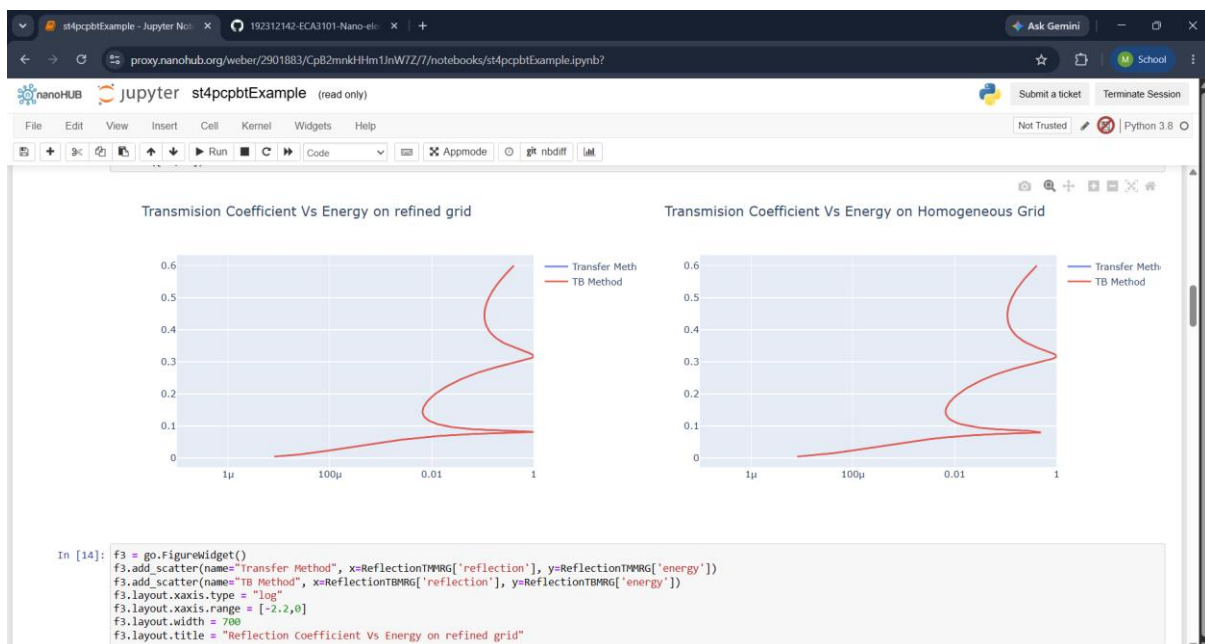


Case Study 2: Simulate the band-structure of AlGaAs circular nanowire

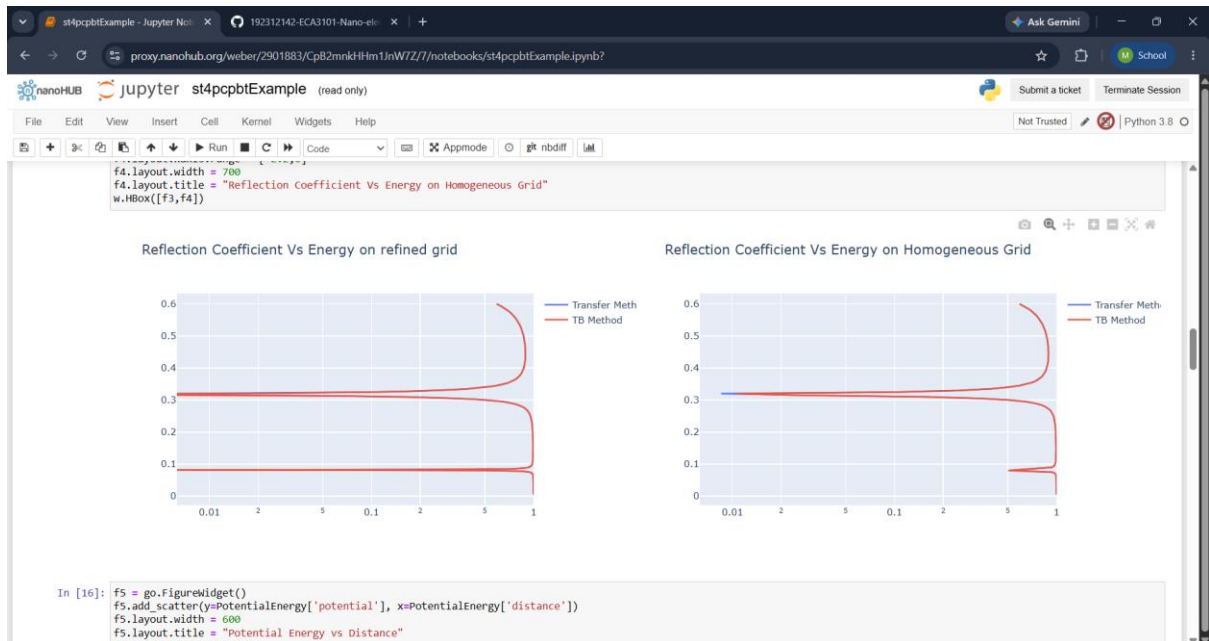


Exp-6 Demonstration of quantum tunnelling

Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid



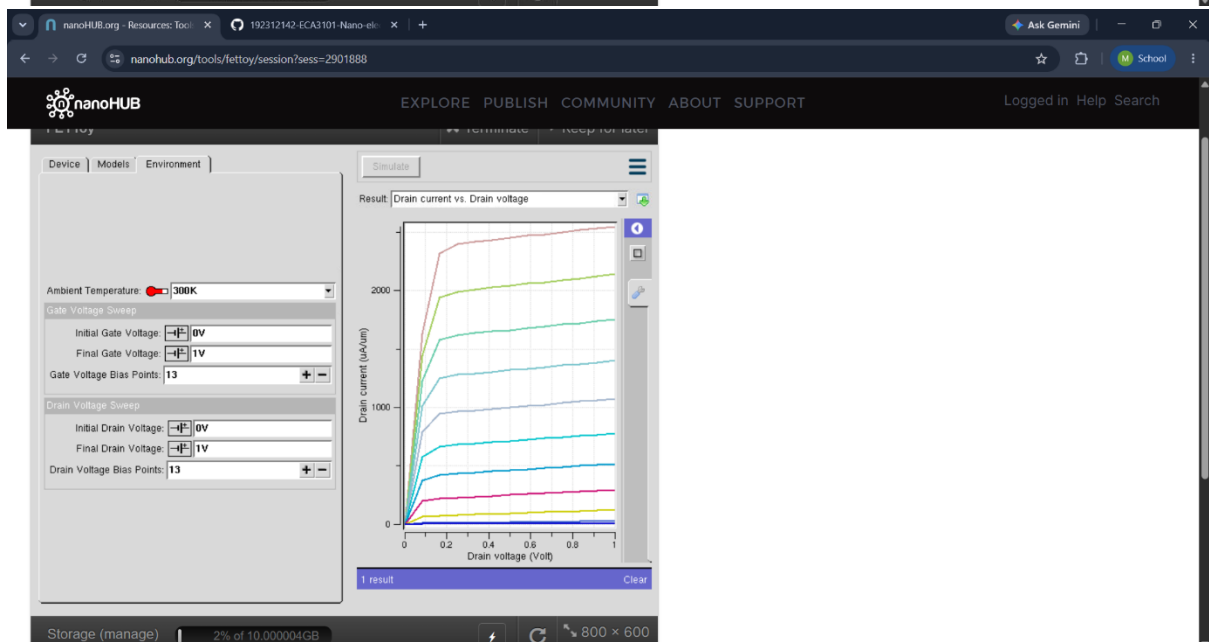
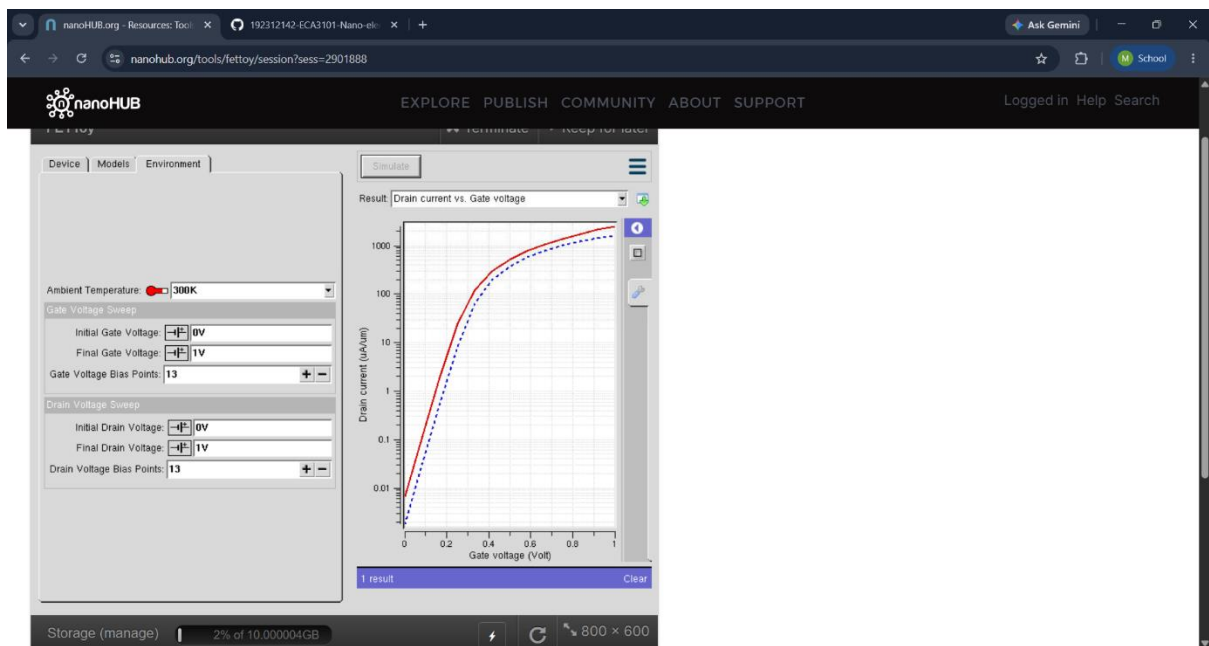
Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid



Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

Case 1: Increase in V_G :

- **Observation:** As V_G increases, the drain current (I_D) also increases exponentially due to the MOSFET entering the saturation region earlier.



Case 2: Variation in V_D :

- **Observation: Drain current (I_D) saturates for higher V_D at a fixed V_G**

